

relative_bias_plots

2026-03-30

```
library(ggplot2)
library(dplyr)

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

options(scipen = 999)

# Load data ----
setwd("/Users/remus/Documents/GitHub/latent_interaction/sim_latent_interaction")
load(file = "agg_results.rda")

# Get unique combinations of rsq_prod and n_items
combinations <- agg_results %>%
  distinct(rsq_prod, n_items)

i=1

# Loop over each combination
for (i in 1:nrow(combinations)) {

  plot_data <- agg_results %>%
    filter(rsq_prod == combinations$rsq_prod[i], n_items == combinations$n_items[i],
           group_prob %in% c(1,2))

  # Skip if no data
  if (nrow(plot_data) == 0) next

  plot_data <- plot_data %>%
    mutate(
      group_prob = factor(group_prob, levels = c(1, 2)),
      loading     = factor(loading),
```

```

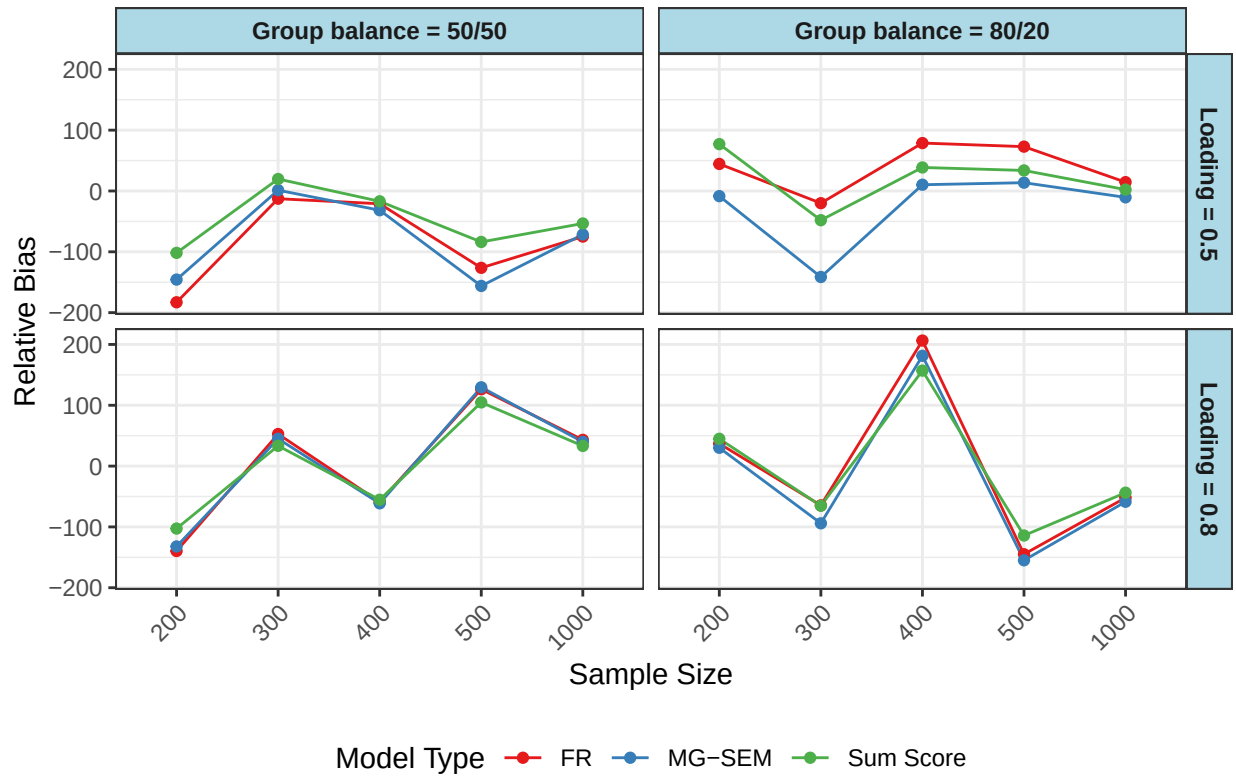
model_type = factor(model_type, levels = c(1, 2, 3),
                    labels = c("FR", "MG-SEM", "Sum Score")),
sample_size = factor(sample_size))

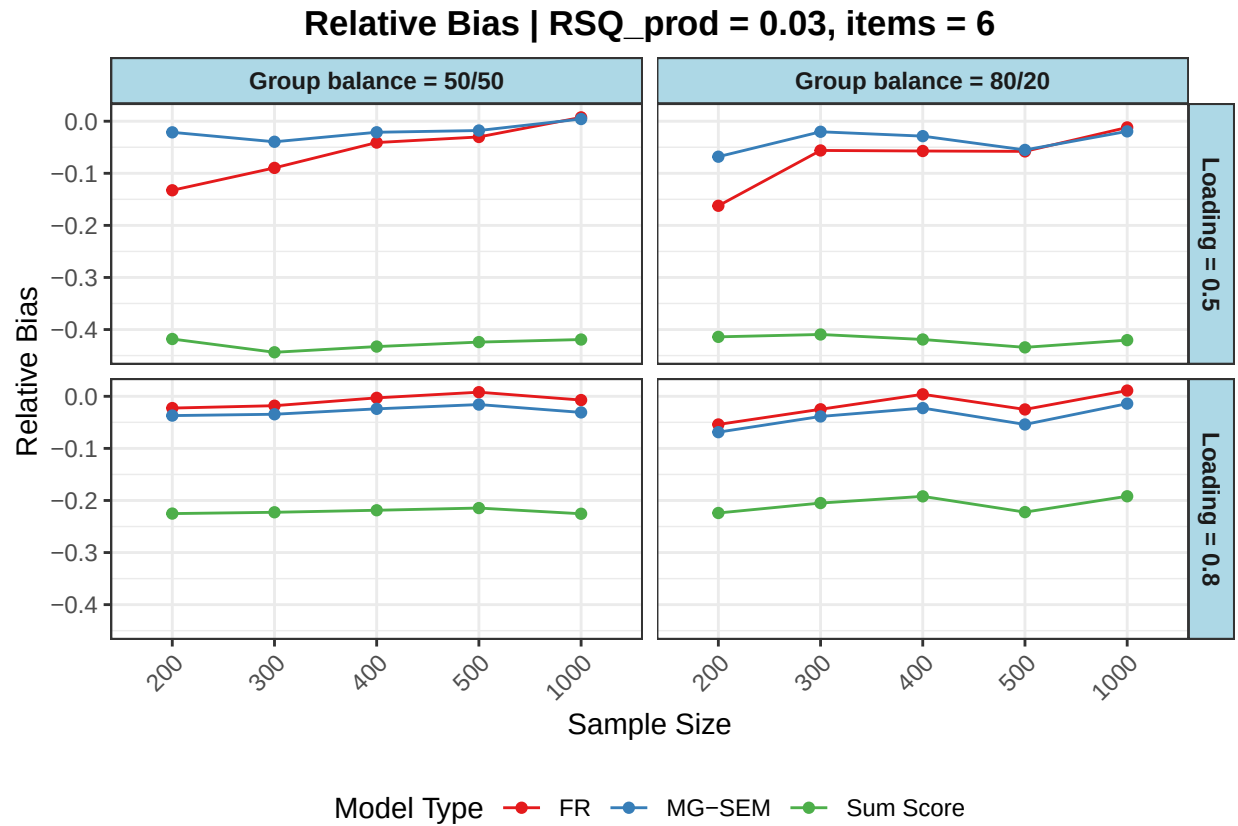
p <- ggplot(plot_data, aes(x = sample_size, y = rel.bias,
                        color = model_type,
                        group = model_type)) +
  geom_line() + geom_point() + facet_grid(rows = vars(loading),
                                           cols = vars(group_prob),
                                           labeller = labeller(
                                             loading = function(x) paste("Loading =", x),
                                             group_prob = function(x) {
                                               x_new <- ifelse(x == 1, "50/50",
                                                             ifelse(x == 2, "80/20", x))
                                               paste("Group balance =", x_new)})) +
  scale_color_brewer(palette = "Set1", name = "Model Type") +
  labs(
    title = paste0("Relative Bias | RSQ_prod = ", combinations$rsq_prod[i],
                  ", items = ", combinations$n_items[i]),
    x = "Sample Size", y = "Relative Bias") +
  theme_bw() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold"),
    legend.position = "bottom",
    axis.text.x = element_text(angle = 45, hjust = 1),
    strip.background = element_rect(fill = "lightblue"),
    strip.text = element_text(face = "bold"))

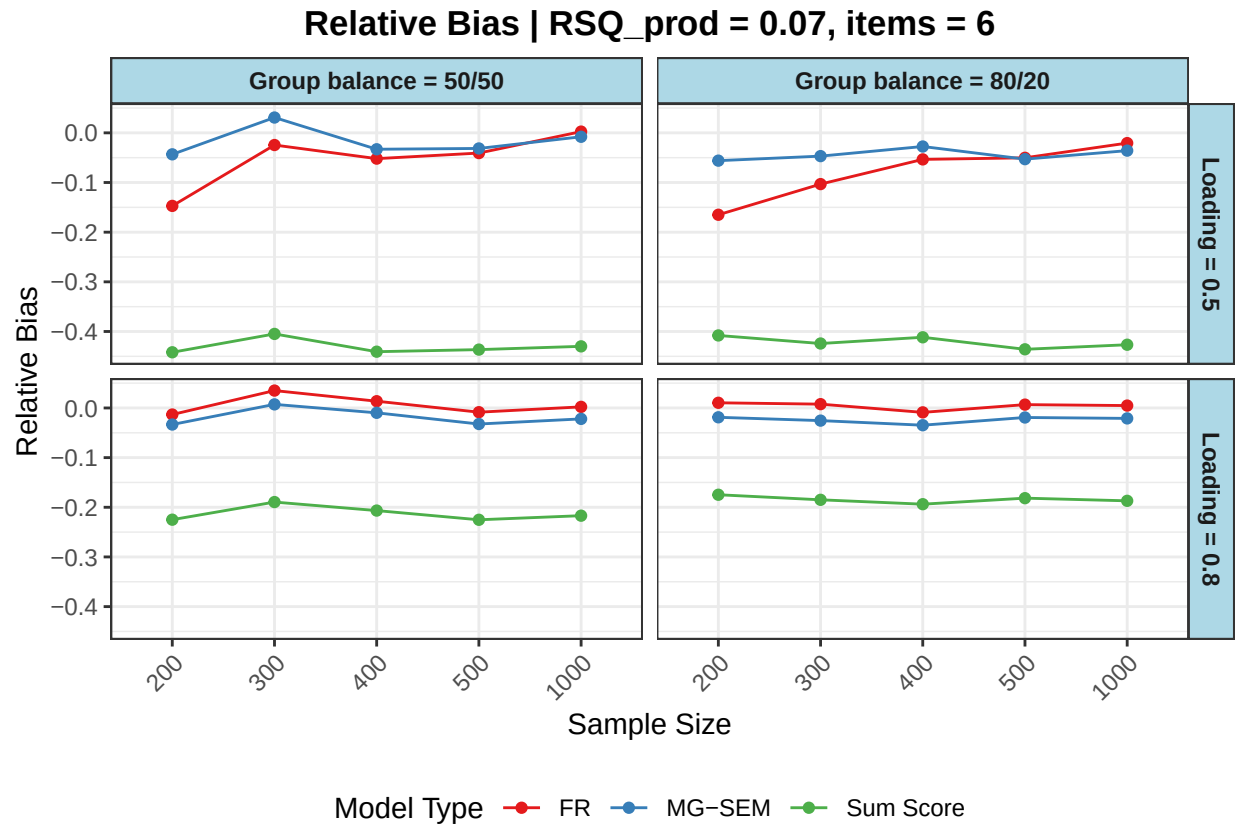
print(p)
}

```

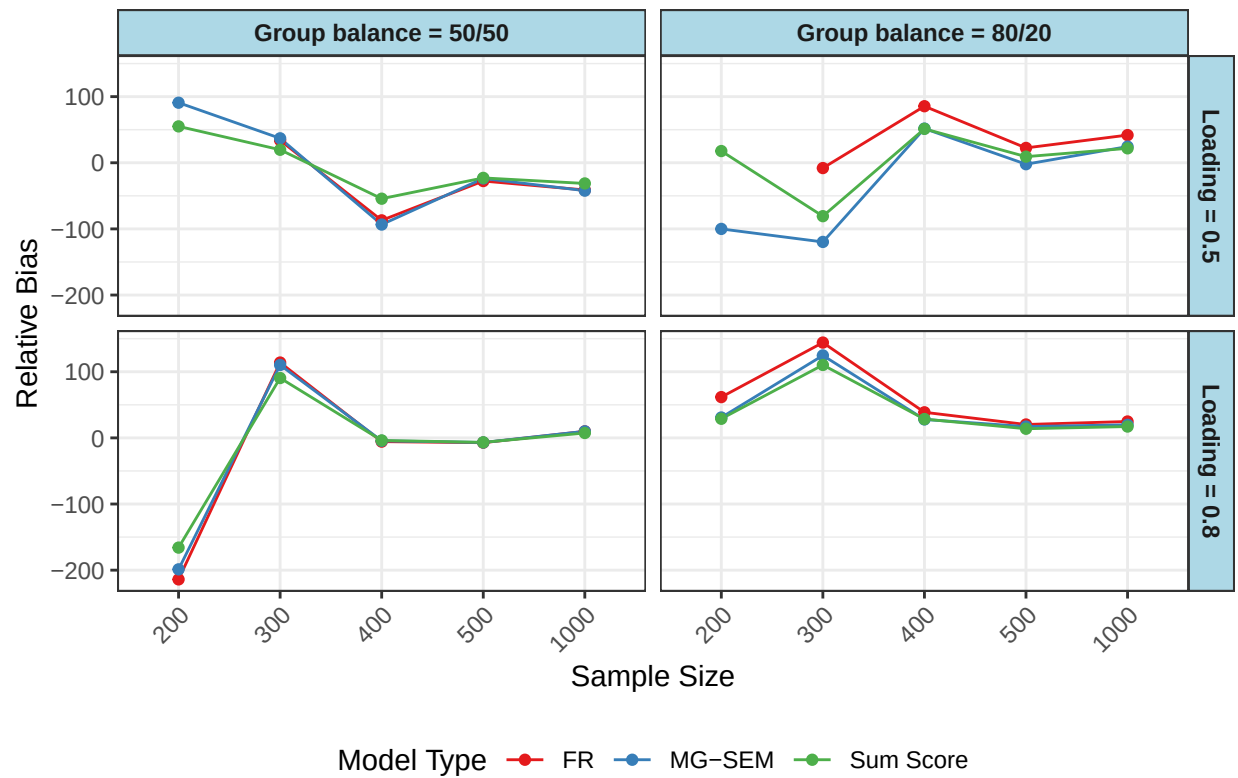
Relative Bias | RSQ_prod = 0, items = 6



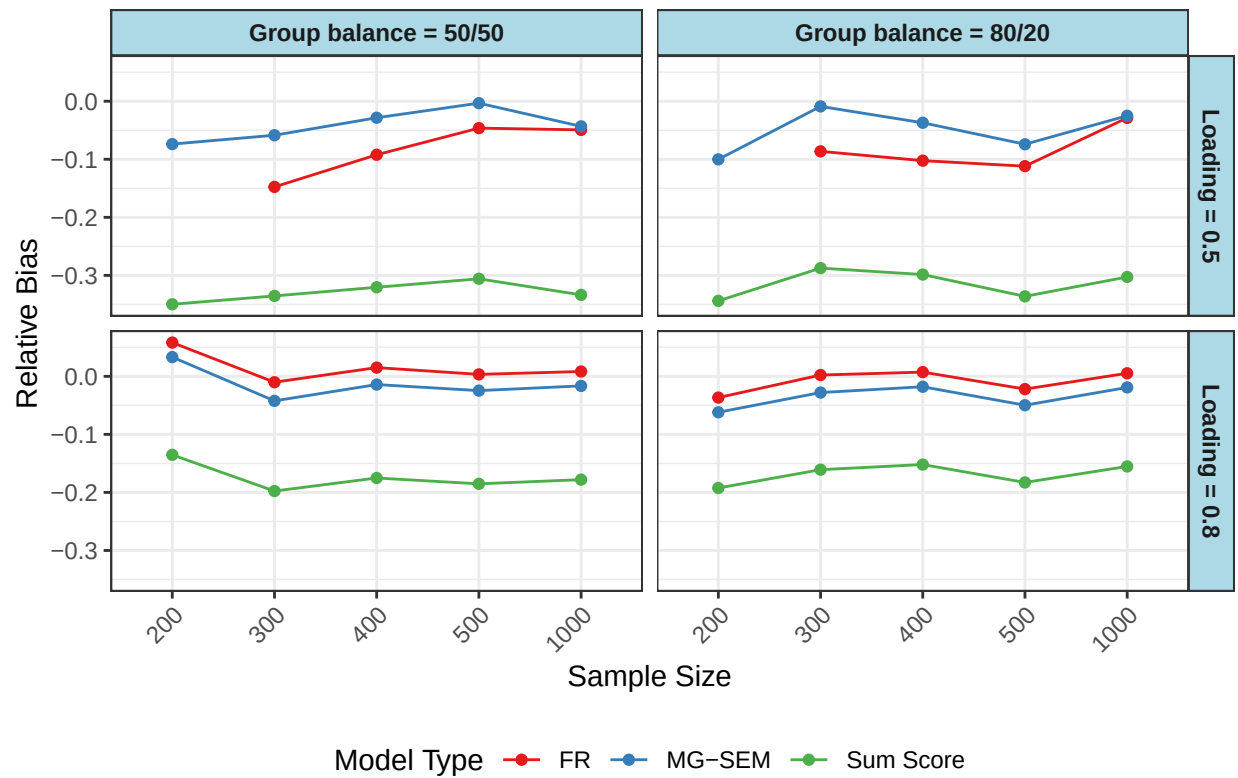




Relative Bias | RSQ_prod = 0, items = 12



Relative Bias | RSQ_prod = 0.03, items = 12



Relative Bias | RSQ_prod = 0.07, items = 12

